第十七讲
应变状态分析
广义胡克定律

作业：8-11(c),12(c),14,18,20,22;

第八章　应力应变状态分析

8.1 平面应力状态应力分析
8.2 应力圆
8.3 极值应力与主应力
8.4 复杂应力状态的最大应力
8.5 平面内应变分析
8.6 广义胡克定律
8.7 复杂应力状态下的应变能
8.8 复合材料的应力应变关系简介（拓展内容）

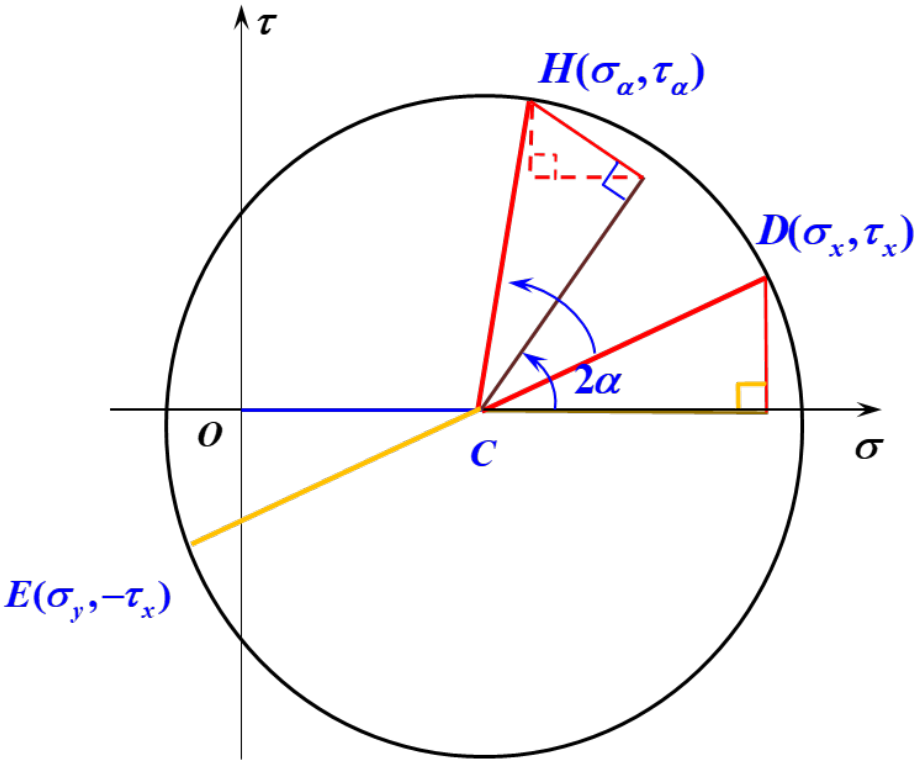
上讲回顾

应力转轴公式

$$\sigma_\alpha = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\alpha - \tau_x \sin 2\alpha$$

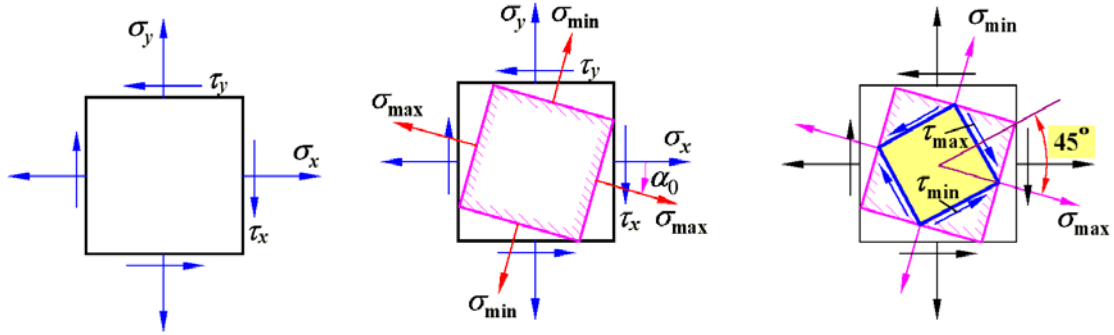
$$\tau_\alpha = \frac{\sigma_x - \sigma_y}{2}\sin 2\alpha + \tau_x \cos 2\alpha$$

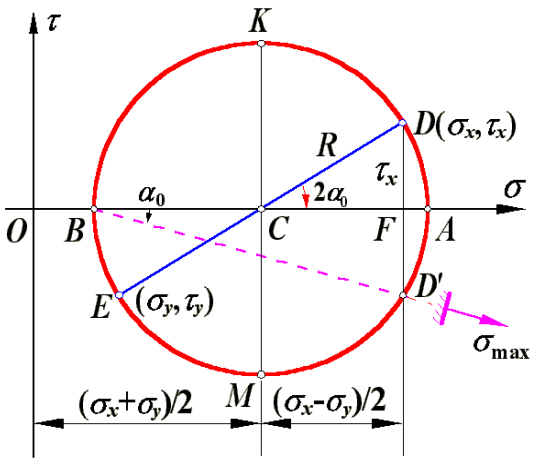
应力圆



上讲回顾

应力圆





$$\begin{matrix}\sigma_{max}\\ \sigma_{min}\end{matrix} = \frac{\sigma_x+\sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x-\sigma_y}{2}\right)^2+\tau_x^2}$$

$$\begin{matrix}\tau_{max}\\ \tau_{min}\end{matrix} = \pm\sqrt{\left(\frac{\sigma_x-\sigma_y}{2}\right)^2+\tau_x^2}$$

$$\tan 2\alpha_0 = -\frac{2\tau_x}{\sigma_x-\sigma_y}$$

$$\tan\alpha_0 = -\frac{\tau_x}{\sigma_x-\sigma_{min}} = -\frac{\tau_x}{\sigma_{max}-\sigma_y}$$

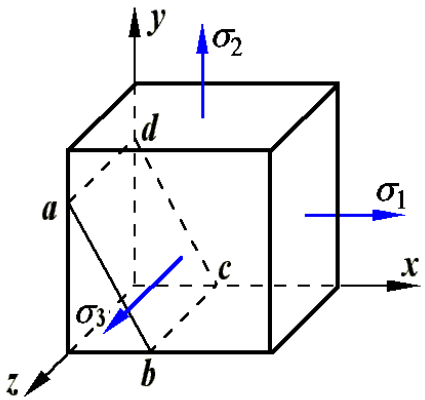
点面对应，转向相同，转角加倍

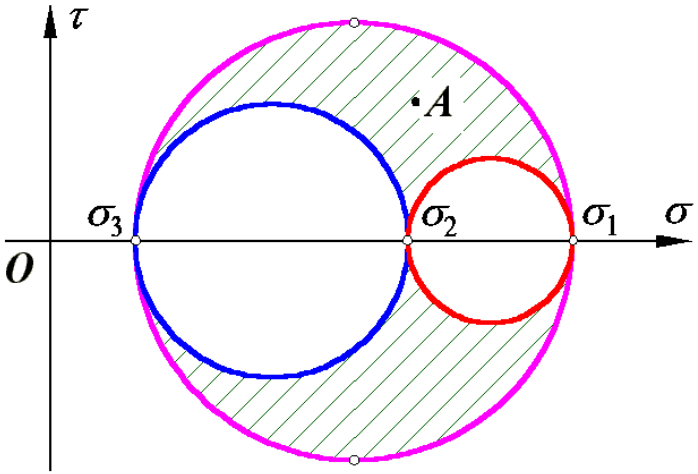
主平面与主应力

$$\sigma_1 \geq \sigma_2 \geq \sigma_3$$

上讲回顾

三向应力圆



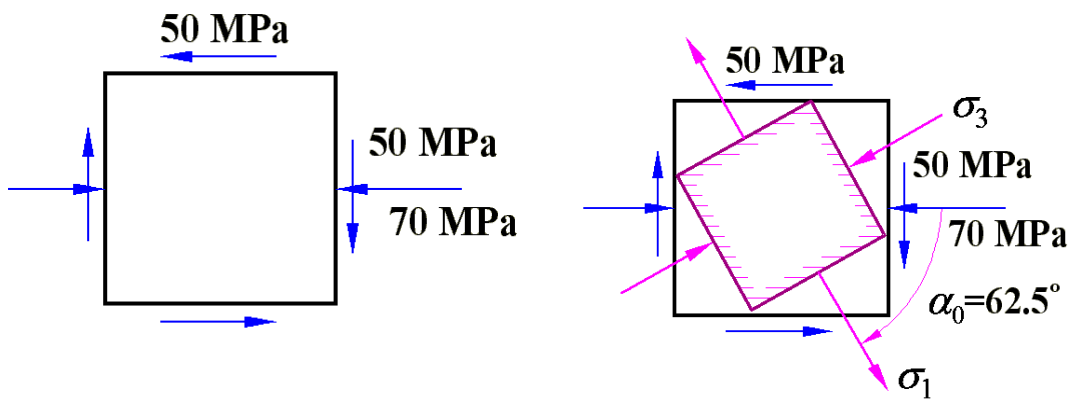


$$\sigma_{max} = \sigma_1 \qquad \sigma_{min} = \sigma_3$$

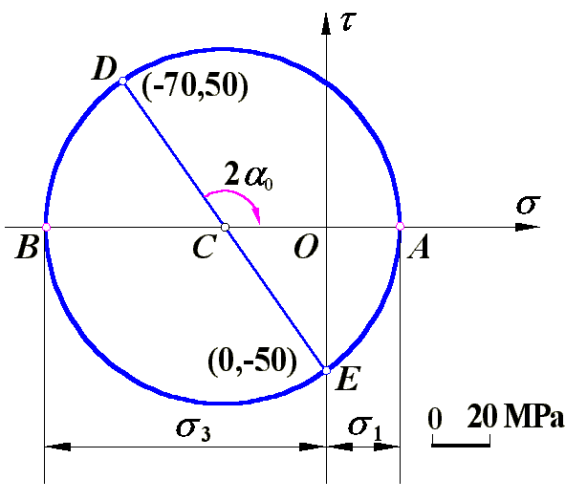
$$\tau_{max} = \frac{\sigma_1 - \sigma_3}{2}$$

上讲回顾

例8-4 用解析法与图解法，确定主应力的大小与方位



图解法



解析法

$$\alpha_0 = \arctan\left(-\frac{50}{26-0}\right) = -62.5^\circ$$

若按公式 $\tan 2\alpha_0 = -\dfrac{2\tau_x}{\sigma_x - \sigma_y}$

$$\alpha_0 = \frac{1}{2}\arctan\left(-\frac{2\times 50}{-70-0}\right) = \frac{1}{2}\times 55.0^\circ = 27.5^\circ$$

为什么？

第十七讲
应变状态分析
广义胡克定律

作业：8-11(c),12(c),14,18,20,22

第八章　应力应变状态分析

8.1 平面应力状态应力分析
8.2 应力圆
8.3 极值应力与主应力
8.4 复杂应力状态的最大应力
8.5 平面内应变分析
8.6 广义胡克定律
8.7 复杂应力状态下的应变能
8.8 复合材料的应力应变关系简介（拓展内容）

第八章 应力应变状态分析

8.5 平面内应变分析

1、任意方位的应变

2、应变圆

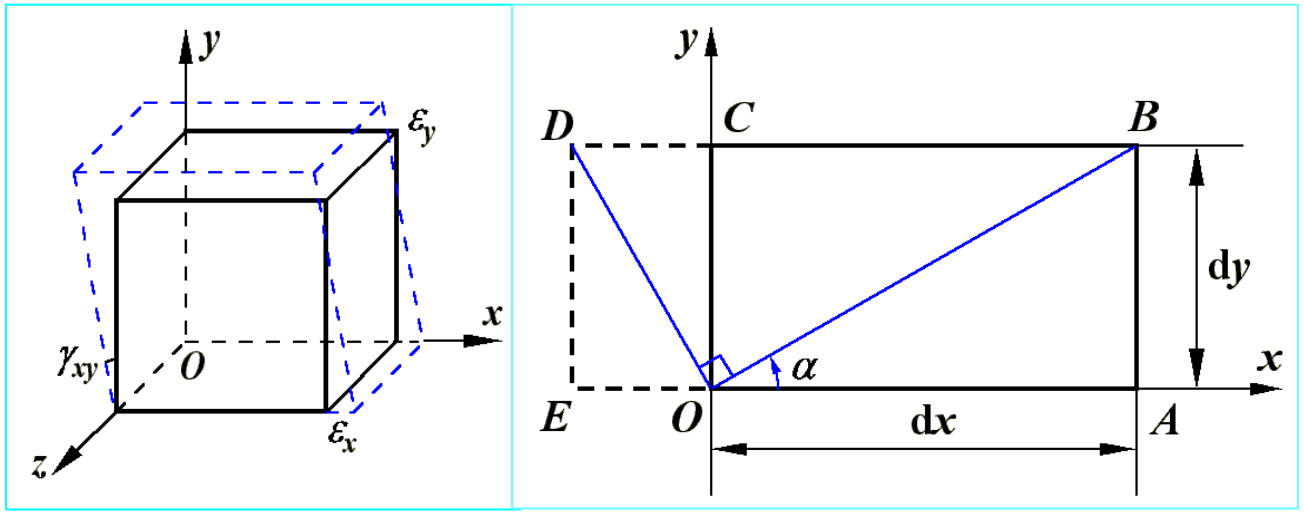
3、最大应变与主应变

4、例题

1、任意方位的应变

问题： 已知应变 ε_x，ε_y与 γ_{xy}，求 α 方位的应变 ε_α 与 γ_α。

平面内变形分析



定义：

1）α方位：**OB**的方位，α角以 x 轴为始边，↺为正；

2）α方位正应变ε_α：**OB**向正应变，拉应变为正；

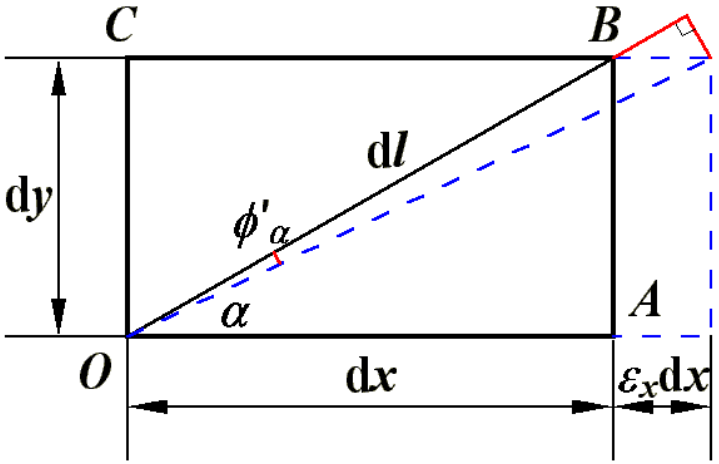
3）α方位切应变γ_α：直角**BOD**的改变量，增大为正。

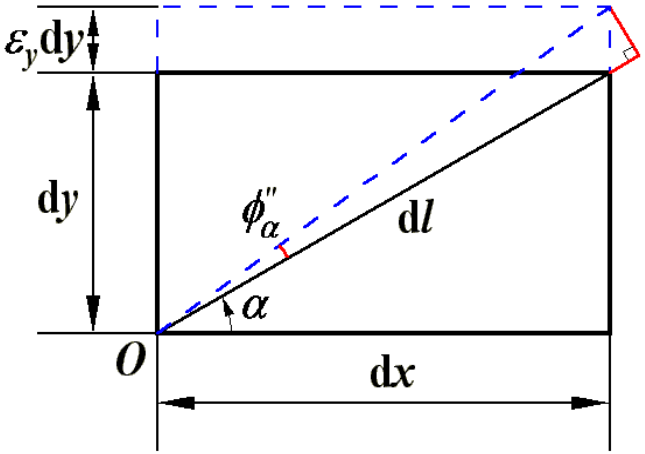
1、任意方位的应变

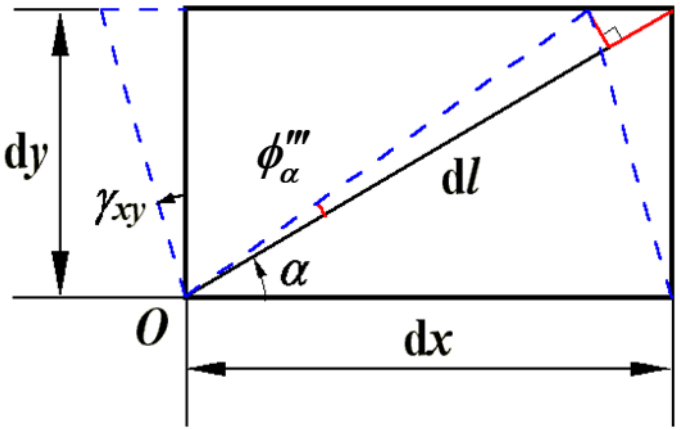
分析方法：

切线代圆弧，

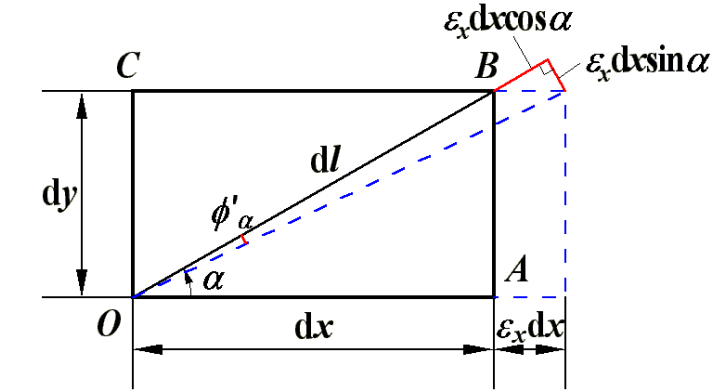
叠加法。

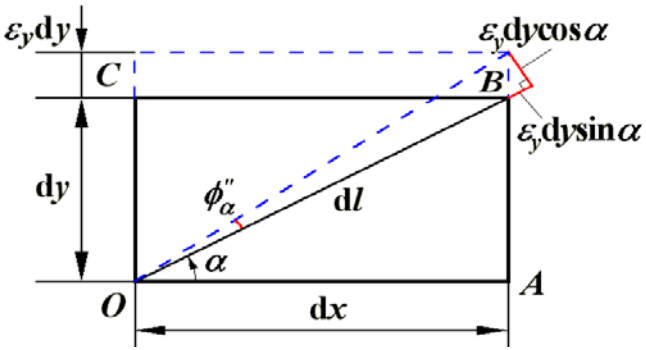


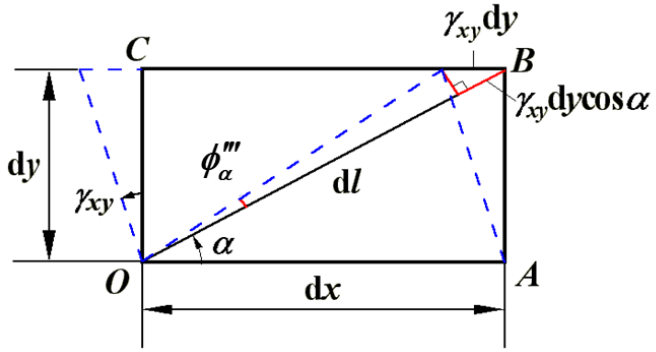




1、任意方位的应变







切线代圆弧

$$\varepsilon'_\alpha = \frac{\varepsilon_x \mathrm{d}x\cos\alpha}{\mathrm{d}l} = \varepsilon_x \cos^2\alpha$$

$$\phi'_\alpha = -\frac{\varepsilon_x \mathrm{d}x\sin\alpha}{\mathrm{d}l} = -\varepsilon_x \cos\alpha\sin\alpha$$

$$\varepsilon''_\alpha = \frac{\varepsilon_y \mathrm{d}y\sin\alpha}{\mathrm{d}l} = \varepsilon_y \sin^2\alpha$$

$$\phi''_\alpha = \frac{\varepsilon_y \mathrm{d}y\cos\alpha}{\mathrm{d}l} = \varepsilon_y \sin\alpha\cos\alpha$$

$$\varepsilon'''_\alpha = -\frac{\gamma_{xy} \mathrm{d}y\cos\alpha}{\mathrm{d}l} = -\frac{\gamma_{xy}\sin 2\alpha}{2}$$

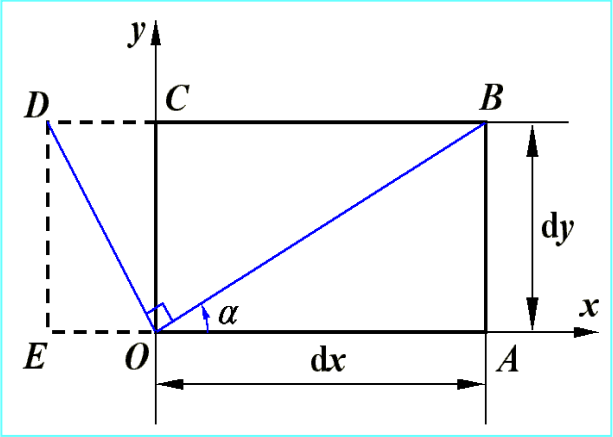
$$\phi'''_\alpha = \frac{\gamma_{xy} \mathrm{d}y\sin\alpha}{\mathrm{d}l} = \gamma_{xy}\sin^2\alpha$$

1、任意方位的应变

应变叠加

$$\varepsilon_\alpha = \varepsilon'_\alpha + \varepsilon''_\alpha + \varepsilon'''_\alpha$$

$$\varepsilon_\alpha = \varepsilon_x \cos^2\alpha + \varepsilon_y \sin^2\alpha - \gamma_{xy}\cos\alpha\sin\alpha$$



转角叠加

$$\phi_\alpha = \phi'_a + \phi''_\alpha + \phi'''_\alpha$$

$$\phi_\alpha = (\varepsilon_y - \varepsilon_x)\cos\alpha\sin\alpha + \gamma_{xy}\sin^2\alpha$$

$$\phi_{\alpha+90^\circ} = (\varepsilon_x - \varepsilon_y)\cos\alpha\sin\alpha + \gamma_{xy}\cos^2\alpha$$

也可取*CDEO*微体，确定 ϕ_{OD}

$$\gamma_\alpha = \phi_{\alpha+90^\circ} - \phi_\alpha \quad \Rightarrow \quad \gamma_\alpha = (\varepsilon_x - \varepsilon_y)\sin 2\alpha + \gamma_{xy}\cos 2\alpha$$

1、任意方位的应变

应变转轴公式

$$\varepsilon_\alpha = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\alpha - \frac{\gamma_{xy}}{2}\sin 2\alpha$$

$$\frac{\gamma_\alpha}{2} = \frac{\varepsilon_x - \varepsilon_y}{2}\sin 2\alpha + \frac{\gamma_{xy}}{2}\cos 2\alpha$$

可见：

1）同一直线上： $\varepsilon_\alpha = \varepsilon_{\alpha+\pi}$， $\gamma_\alpha = \gamma_{\alpha+\pi}$

2）互相垂直方位切应变：

$\gamma_\alpha = -\gamma_{\alpha+90^\circ}$ 数值相等、符号相反

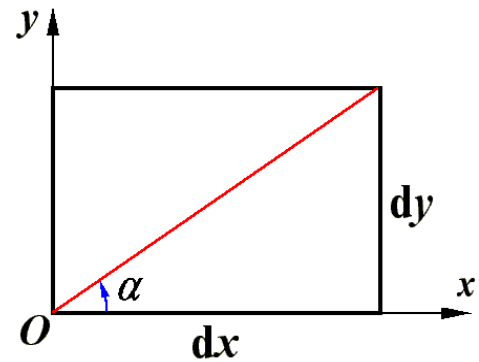
上述分析建立在几何关系基础上，所得结论适用于任何小变形问题，而与材料的力学特性无关

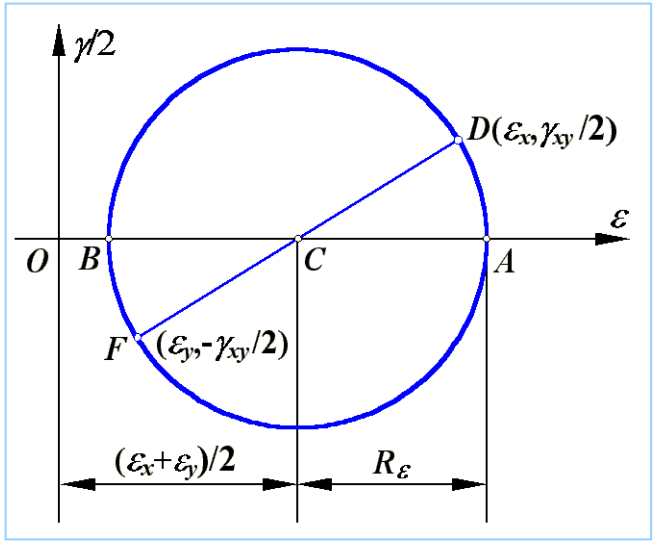
2、应变圆

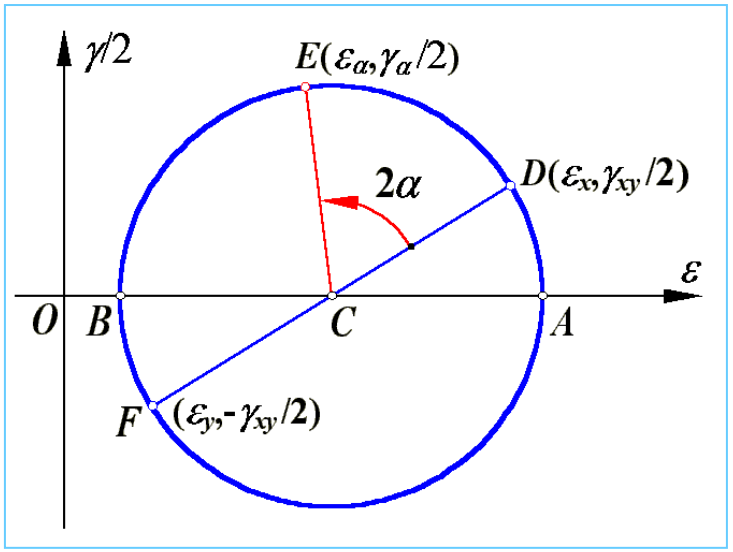
$$\varepsilon_\alpha = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\alpha - \frac{\gamma_{xy}}{2}\sin 2\alpha$$

$$\frac{\gamma_\alpha}{2} = \frac{\varepsilon_x - \varepsilon_y}{2}\sin 2\alpha + \frac{\gamma_{xy}}{2}\cos 2\alpha$$

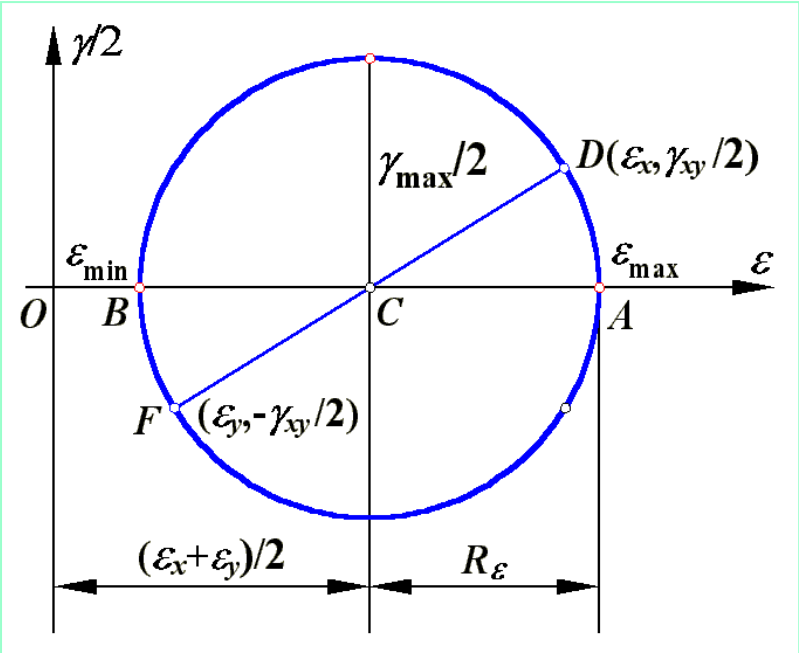
$$\left(\varepsilon_\alpha - \frac{\varepsilon_x + \varepsilon_y}{2}\right)^2 + \left(\frac{\gamma_\alpha}{2} - 0\right)^2 = \left(\frac{\varepsilon_x - \varepsilon_y}{2}\right)^2 + \left(\frac{\gamma_{xy}}{2}\right)^2 = R_\varepsilon^2$$

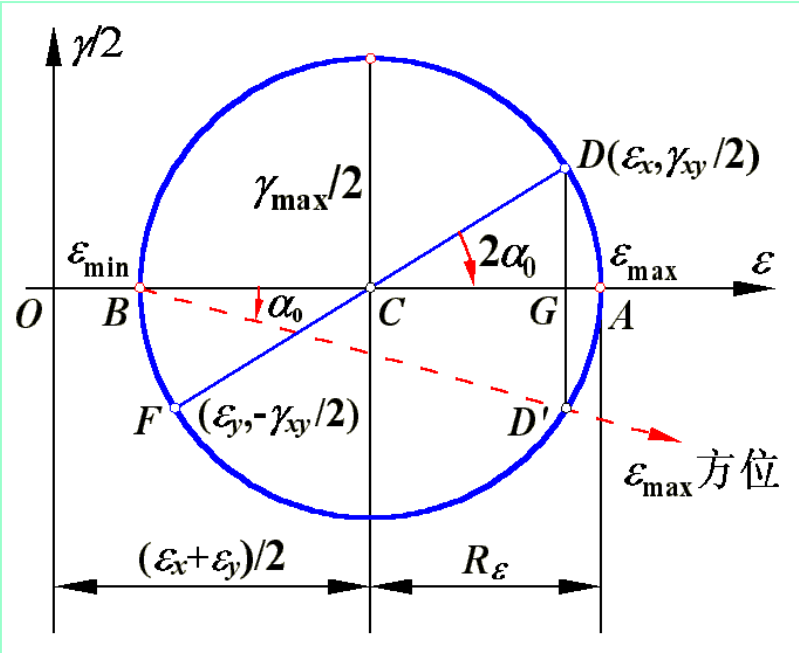






3、最大应变与主应变



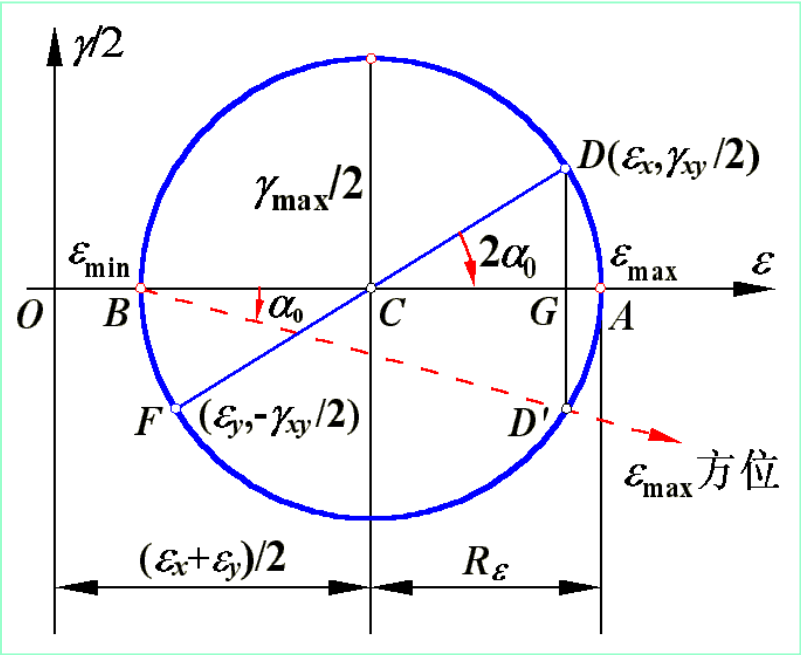


$$\begin{matrix}\varepsilon_{\max}\\ \varepsilon_{\min}\end{matrix}=\frac{\varepsilon_x+\varepsilon_y}{2}\pm\frac{1}{2}\sqrt{\left(\varepsilon_x-\varepsilon_y\right)^2+\gamma_{xy}^2}$$

$$\gamma_{\max}=\sqrt{\left(\varepsilon_x-\varepsilon_y\right)^2+\gamma_{xy}^2}$$

$$\tan\alpha_0=-\frac{\gamma_{xy}/2}{\varepsilon_x-\varepsilon_{\min}}=-\frac{\gamma_{xy}}{2\left(\varepsilon_x-\varepsilon_{\min}\right)}$$

3、最大应变与主应变



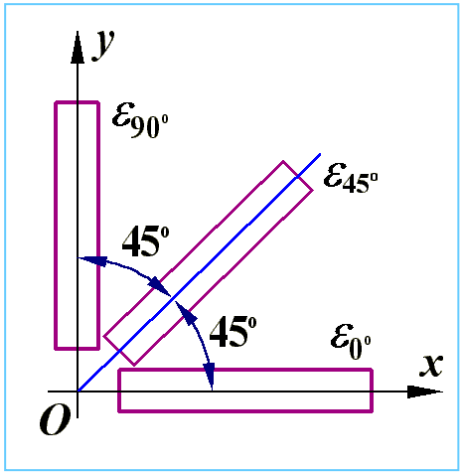
切应变为零方位的正应变—**主应变**

主应变位于互垂方位

主应变表示：$\varepsilon_1 \geq \varepsilon_2 \geq \varepsilon_3$

4、例题

例8-5　构件表面一点处粘贴图示应变花，由试验测得0°，45°与 90°方位的应变分别为ε_0，ε_{45}与ε_{90}，求 ε_x，ε_y与 γ_{xy}



解：

$$\varepsilon_\alpha = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\alpha - \frac{\gamma_{xy}}{2}\sin 2\alpha$$

$$\varepsilon_{0°} = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\cdot 0° - \frac{\gamma_{xy}}{2}\sin 2\cdot 0°$$

$$\varepsilon_{45°} = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\cdot 45° - \frac{\gamma_{xy}}{2}\sin 2\cdot 45°$$

$$\varepsilon_{90°} = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\cdot 90° - \frac{\gamma_{xy}}{2}\sin 2\cdot 90°$$

$$\varepsilon_x = \varepsilon_{0°} \qquad \varepsilon_y = \varepsilon_{90°} \qquad \gamma_{xy} = 2\varepsilon_{45°} - (\varepsilon_{0°} + \varepsilon_{90°})$$

第八章 应力应变状态分析

8.6 广义胡克定律

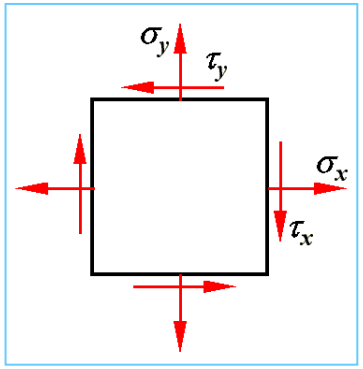
1、广义胡克定律

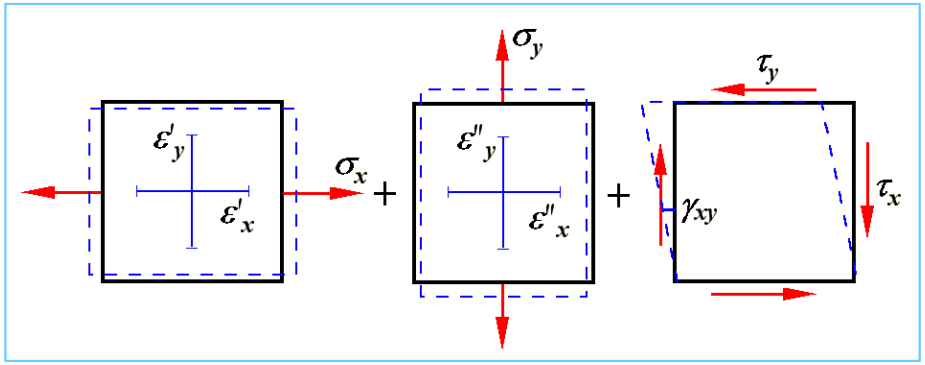
2、弹性常数E、G、μ之关系

3、例题

1、广义胡克定律

平面应力状态





$$\varepsilon'_x = \frac{\sigma_x}{E} \qquad \varepsilon'_y = -\frac{\mu\sigma_x}{E} \qquad \varepsilon''_y = \frac{\sigma_y}{E} \qquad \varepsilon''_x = -\frac{\mu\sigma_y}{E}$$

$$\varepsilon_x = \frac{1}{E}(\sigma_x - \mu\sigma_y) \qquad \sigma_x = \frac{E}{1-\mu^2}(\varepsilon_x + \mu\varepsilon_y)$$

$$\varepsilon_y = \frac{1}{E}(\sigma_y - \mu\sigma_x) \qquad \sigma_y = \frac{E}{1-\mu^2}(\varepsilon_y + \mu\varepsilon_x)$$

$$\gamma_{xy} = \frac{\tau_x}{G} = \frac{2(1+\mu)}{E}\tau_x \qquad \tau_{xy} = G\gamma_{xy} = \frac{E}{2(1+\mu)}\gamma_{xy}$$

适用范围：各向同性材料，线弹性范围内

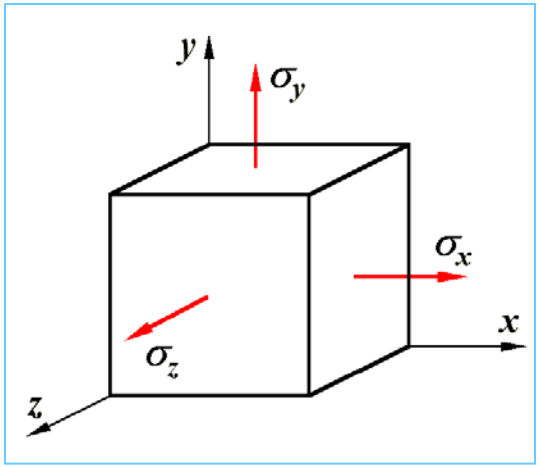
1、广义胡克定律

三向应力状态

$$\varepsilon_x = \frac{1}{E}[\sigma_x - \mu(\sigma_y + \sigma_z)]$$

$$\varepsilon_y = \frac{1}{E}[\sigma_y - \mu(\sigma_z + \sigma_x)]$$

$$\varepsilon_z = \frac{1}{E}[\sigma_z - \mu(\sigma_x + \sigma_y)]$$



适用范围：各向同性材料，线弹性范围内

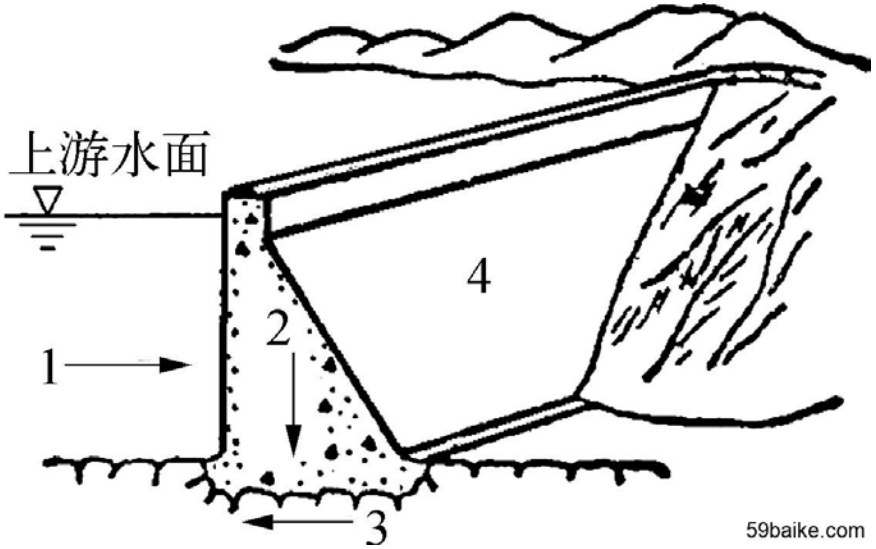
1、广义胡克定律

平面应变状态

构件内每一点处的应变均在同一平面内

$$\varepsilon_z = \frac{1}{E}[\sigma_z - \mu(\sigma_x + \sigma_y)] = 0$$

$$\sigma_z = \mu(\sigma_x + \sigma_y)$$



$$\varepsilon_x = \frac{1-\mu^2}{E}\left(\sigma_x - \frac{\mu}{1-\mu}\sigma_y\right)$$

$$\varepsilon_y = \frac{1-\mu^2}{E}\left(\sigma_y - \frac{\mu}{1-\mu}\sigma_x\right)$$

$$\gamma_{xy} = \frac{\tau_x}{G} = \frac{2(1+\mu)}{E}\tau_x$$

$$E \to \frac{E}{1-\mu^2}, \quad \mu \to \frac{\mu}{1-\mu}$$

平面应力→平面应变

1、广义胡克定律

主应力与主应变的关系

1）主应变与主应力的方位重合

2）最大、最小主应变分别发生在最大、最小主应力方位

$$\varepsilon_1=\frac{1}{E}[\sigma_1-\mu(\sigma_2+\sigma_3)]=\frac{1}{E}[(1+\mu)\sigma_1-\mu(\sigma_1+\sigma_2+\sigma_3)]$$

$$\varepsilon_2=\frac{1}{E}[\sigma_2-\mu(\sigma_3+\sigma_1)]=\frac{1}{E}[(1+\mu)\sigma_2-\mu(\sigma_1+\sigma_2+\sigma_3)]$$

$$\varepsilon_3=\frac{1}{E}[\sigma_3-\mu(\sigma_1+\sigma_2)]=\frac{1}{E}[(1+\mu)\sigma_3-\mu(\sigma_1+\sigma_2+\sigma_3)]$$

$$\therefore\ \varepsilon_1\geq\varepsilon_2\geq\varepsilon_3$$

3）最大拉应变发生在最大拉应力方位

如果 $\sigma_1\geq 0$，且因 $\mu<1/2$，则 $\varepsilon_{\max}=\varepsilon_1=\frac{1}{E}[\sigma_1-\mu(\sigma_2+\sigma_3)]\geq 0$

2、弹性常数E、G、μ之关系

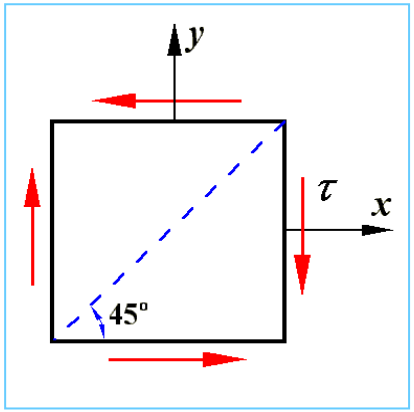
$$\boxed{G=\frac{E}{2(1+\mu)}}$$

证： ❶ 根据几何关系求$\varepsilon_{45°}$

$$\varepsilon_\alpha=\frac{\varepsilon_x+\varepsilon_y}{2}+\frac{\varepsilon_x-\varepsilon_y}{2}\cos 2\alpha-\frac{\gamma_{xy}}{2}\sin 2\alpha$$

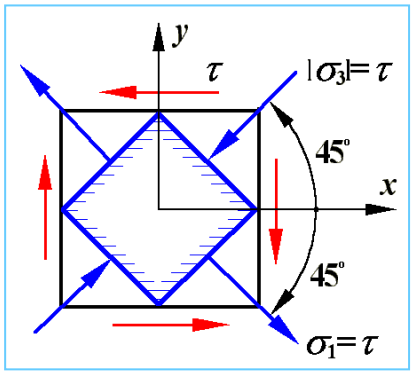
$$\varepsilon_x=\varepsilon_y=0 \qquad \gamma_{xy}=\tau/G$$

$$\varepsilon_{45°}=-\frac{\gamma_{xy}}{2}=-\frac{\tau}{2G}$$



❷ 根据广义胡克定律求 $\varepsilon_{45°}$

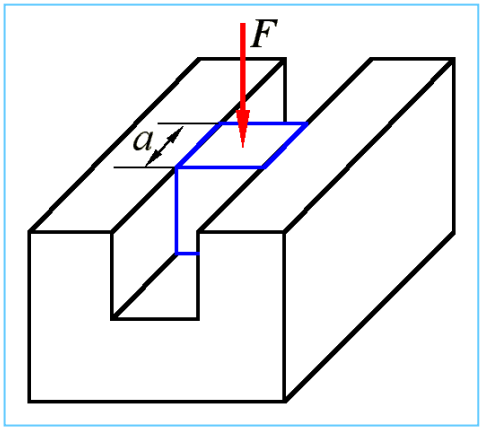
$$\varepsilon_{45°}=\frac{1}{E}(\sigma_3-\mu\sigma_1)=-\frac{(1+\mu)\tau}{E}$$



❸ 比较 $\quad\therefore\ G=\dfrac{E}{2(1+\mu)}$

3、例题

例8-6　边长为 a=10 mm 的正方形钢块放置在槽形刚体内，$F=8\,\text{kN}$，$E=200\,\text{GPa}$，$\mu=0.3$，求：钢块的主应力与主应变。



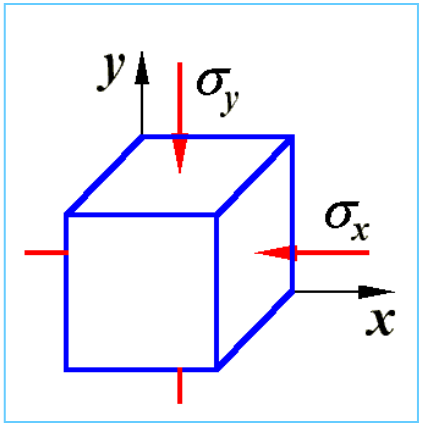
解：$\sigma_y=-\dfrac{F}{a^2}=-80\ \text{MPa}$，$\varepsilon_x=0$，$\sigma_z=0$

$$\varepsilon_x=\frac{\sigma_x}{E}-\frac{\mu\sigma_y}{E}\quad\Rightarrow\quad\sigma_x=\mu\sigma_y=-24\ \text{MPa}$$

$$\sigma_1=0,\quad\sigma_2=-24\ MPa,\quad\sigma_3=-80\ MPa$$

$$\varepsilon_z=0-\frac{\mu\left(\sigma_x+\sigma_y\right)}{E}=156\ \mu\varepsilon,\quad\varepsilon_y=\frac{\sigma_y-\mu\sigma_x}{E}=-364\ \mu\varepsilon$$

$$\varepsilon_1=156\ \mu\varepsilon,\quad\varepsilon_2=0,\quad\varepsilon_3=-364\ \mu\varepsilon$$



受力无变形，无力有变形

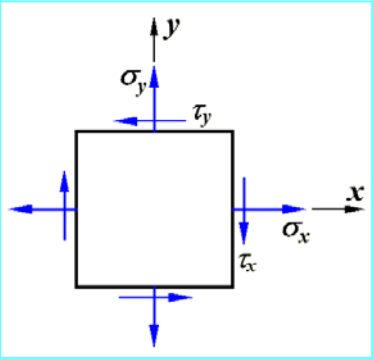
用广义胡克定律描述应力应变关系

❐ 应力应变状态分析图

载荷

横截面
应力分析

危险点
应力状态
是核心



$\sigma_x, \sigma_y, \tau_x$ ⟺ $\varepsilon_x, \varepsilon_y, \gamma_{xy}$ ⟵ 应变花

应力圆 ↓ ↓ 应变圆

胡克定律

$$\varepsilon_\alpha = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\alpha - \frac{\gamma_{xy}}{2}\sin 2\alpha$$

$\sigma_\alpha, \tau_\alpha, \alpha$
$\sigma_1, \sigma_2, \sigma_3$

⟺

$\varepsilon_\alpha, \gamma_\alpha, \alpha$
$\varepsilon_1, \varepsilon_2, \varepsilon_3$

第八章 应力应变状态分析

8.7 复杂应力状态下的应变能

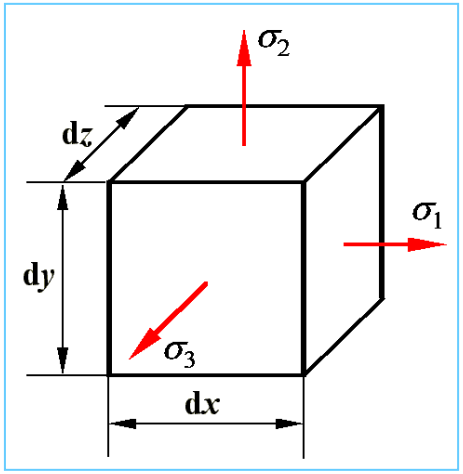
1、应变能密度

2、体积应变与平均应力

3、畸变能密度

1、应变能密度

$$\mathrm{d}W=\mathrm{d}V_{\varepsilon}=\frac{\sigma_1\mathrm{d}y\mathrm{d}z\cdot\varepsilon_1\mathrm{d}x}{2}+\frac{\sigma_2\mathrm{d}z\mathrm{d}x\cdot\varepsilon_2\mathrm{d}y}{2}+\frac{\sigma_3\mathrm{d}x\mathrm{d}y\cdot\varepsilon_3\mathrm{d}z}{2}$$



$$\mathrm{d}V_{\varepsilon}=\frac{1}{2}(\sigma_1\varepsilon_1+\sigma_2\varepsilon_2+\sigma_3\varepsilon_3)\mathrm{d}x\mathrm{d}y\mathrm{d}z$$

单位体积内的应变能 - **应变能密度**

$$v_{\varepsilon}=\frac{1}{2}(\sigma_1\varepsilon_1+\sigma_2\varepsilon_2+\sigma_3\varepsilon_3)$$

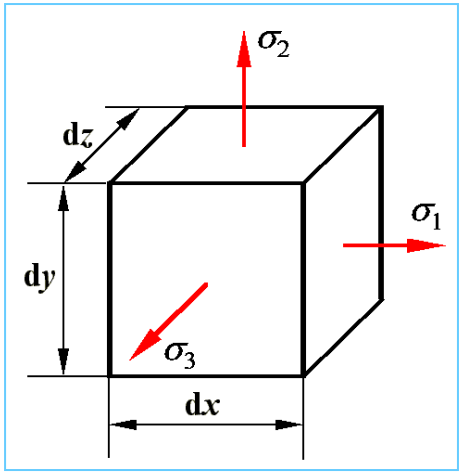
对各向同性材料

$$v_{\varepsilon}=\frac{1}{2E}\left[\sigma_1^2+\sigma_2^2+\sigma_3^2-2\mu(\sigma_1\sigma_2+\sigma_2\sigma_3+\sigma_3\sigma_1)\right]$$

2、体积应变与体积应力

$$dV=(1+\varepsilon_1)(1+\varepsilon_2)(1+\varepsilon_3)dxdydz$$

$$dV\approx(1+\varepsilon_1+\varepsilon_2+\varepsilon_3)dV_o \quad dV_o=dxdydz$$ —原体积



微体的体积变化率 - 体积应变

$$\theta=\frac{dV-dV_o}{dV_o}=\varepsilon_1+\varepsilon_2+\varepsilon_3=\frac{1-2\mu}{E}(\sigma_1+\sigma_2+\sigma_3)$$

$\Theta=\sigma_1+\sigma_2+\sigma_3$ — 体积应力， $\sigma_{av}=\frac{\Theta}{3}$ — 平均应力

$$\boxed{\theta=\frac{(1-2\mu)}{E}\Theta}$$

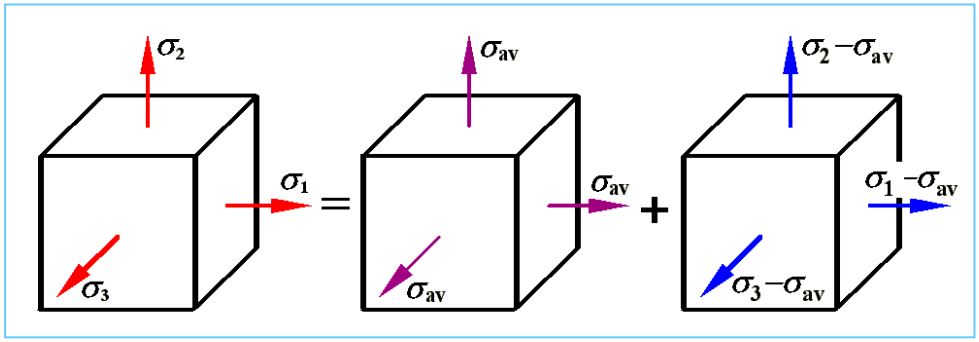
$\frac{E}{1-2\mu}$ —体积弹性模量

μ=0.5，体应变为零，材料不可压缩。

3、畸变能密度

应变能密度

$$v_\varepsilon = \frac{1}{2E}\left[\sigma_1^2+\sigma_2^2+\sigma_3^2 - 2\mu\left(\sigma_1\sigma_2+\sigma_2\sigma_3+\sigma_3\sigma_1\right)\right]$$



体积改变
形状不变

形状改变
体积不变

体积改变能密度

$$v_V = \frac{1-2\mu}{6E}\left(\sigma_1+\sigma_2+\sigma_3\right)^2$$

畸变能密度
（形状改变比能）

$$v_d = \frac{1+\mu}{6E}\left[\left(\sigma_1-\sigma_2\right)^2+\left(\sigma_2-\sigma_3\right)^2+\left(\sigma_3-\sigma_1\right)^2\right]$$

为保证应变能为正

$$-1 \le \mu \le 0.5$$

第八章 应力应变状态分析

8.8 复合材料应力应变关系简介

1、复合材料基本概念

2、单层板正轴应力应变关系

3、偏轴力学特性

1、复合材料基本概念

1）复合材料：

由两种或两种以上性能不同的材料所构成的材料。

2）分类：

- 纤维增强（树脂）复合材料：层合、编织、缝纫；
- 颗粒增强
- 薄片增强

3）单层板：

单向纤维在树脂基体中呈扁平形式的层片——预浸带。

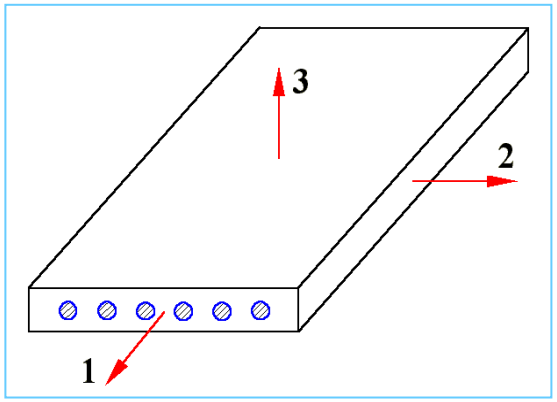
4）层合板壳：

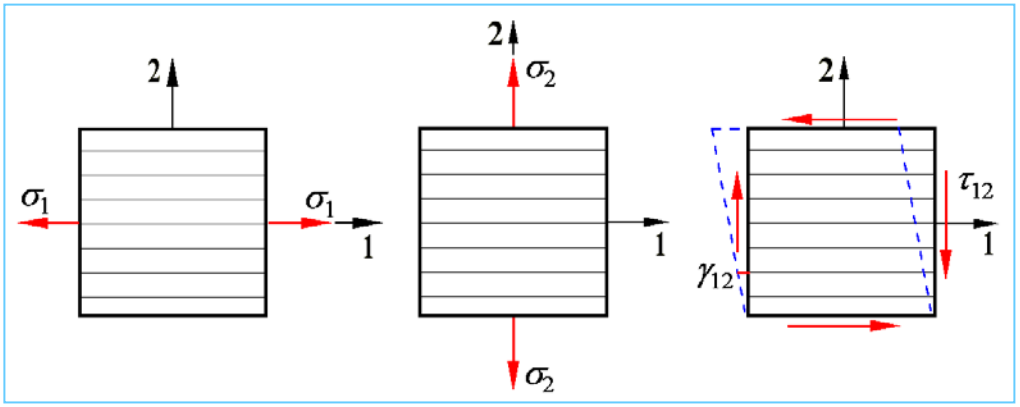
单层板以不同角度铺设后经高温固化而成的板或壳。

5）特点：

比强度、比刚度大，可设计，等等。

2、单层板正轴应力应变关系





$$\varepsilon_{1,\sigma_1}=\frac{\sigma_1}{E_1} \qquad \varepsilon_{2,\sigma_1}=-\mu_{12}\frac{\sigma_1}{E_1}$$

$$\varepsilon_{2,\sigma_2}=\frac{\sigma_2}{E_2} \qquad \varepsilon_{1,\sigma_2}=-\mu_{21}\frac{\sigma_2}{E_2}$$

E_1—纵向弹性模量

μ_{12}—纵向泊松比

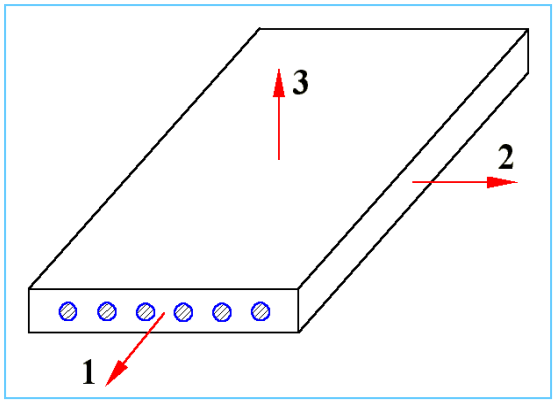
E_2—横向弹性模量

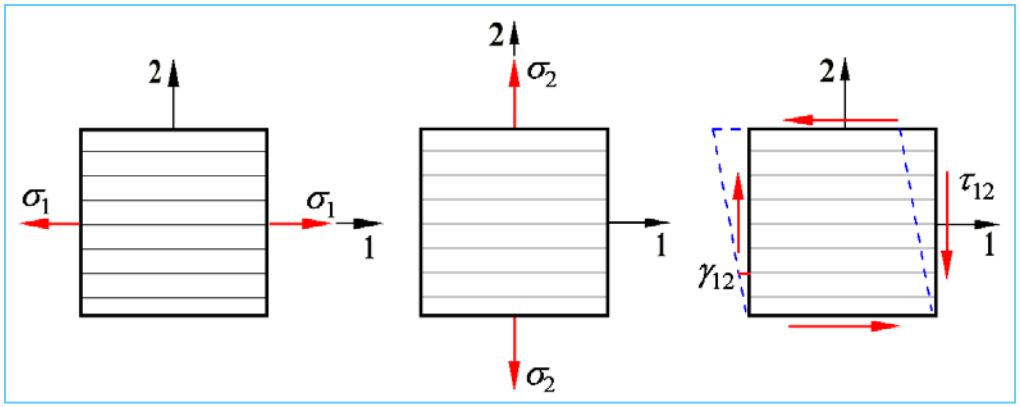
μ_{21}—横向泊松比

$$\varepsilon_1=\frac{\sigma_1}{E_1}-\frac{\mu_{21}\sigma_2}{E_2} \qquad \varepsilon_2=\frac{\sigma_2}{E_2}-\frac{\mu_{12}\sigma_1}{E_1} \qquad \gamma_{12}=\frac{\tau_{12}}{G_{12}}$$

G_{12}—纵向切变模量

2、单层板正轴应力应变关系



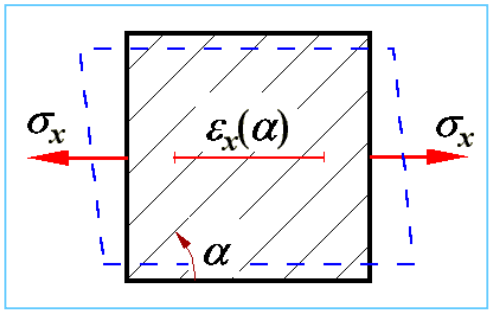


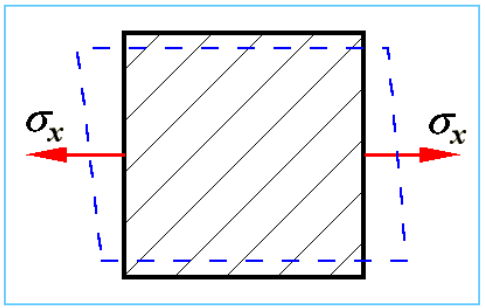
$$\begin{Bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \gamma_{12} \end{Bmatrix} = \begin{bmatrix} \dfrac{1}{E_1} & -\dfrac{\mu_{21}}{E_2} & 0 \\ -\dfrac{\mu_{12}}{E_1} & \dfrac{1}{E_2} & 0 \\ 0 & 0 & \dfrac{1}{G_{12}} \end{bmatrix} \begin{Bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{Bmatrix}$$

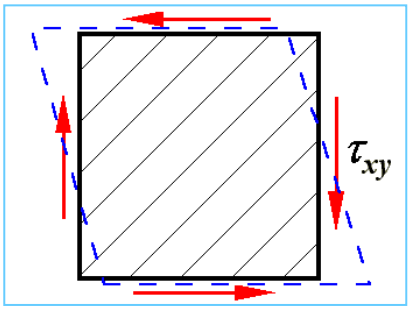
利用功的互等定理可以证明

$$\frac{\mu_{12}}{E_1} = \frac{\mu_{21}}{E_2}$$

3、单层板偏轴力学特性



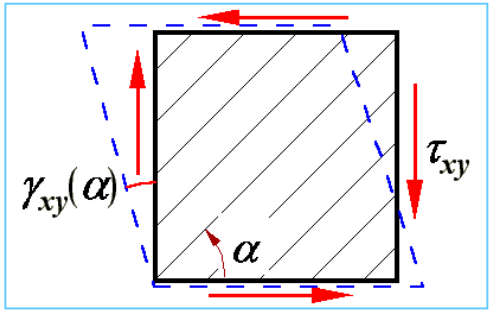




❶ 拉伸与剪切之间存在耦合效应

❷ 应力主轴与应变主轴不重合

❸ 弹性常数具有方向性



$$E_\alpha = \frac{\sigma_x}{\varepsilon_x(\alpha)} = E_\alpha(\alpha)$$

$$G_\alpha = \frac{\tau_{xy}}{\gamma_{xy}(\alpha)} = G_\alpha(\alpha)$$

本讲：应力状态分析

切线代圆弧

下讲：强度理论和薄壁容器